

Setting Up Semihosting in STM32CubeIDE

1 Creating a New Project

1. Launch **STM32CubeIDE**.
2. Create a new project using **File** $\rightarrow$ **New** $\rightarrow$ **STM32 Project**.
3. Select your STM32 microcontroller or board (e.g., **STM32F401RETx**).
4. In the **Project Setup** window, name your project, and select **C** as the language (you can switch to C++ later if needed).
5. Select the option to **Initialize all peripherals with default settings**.

2 Enable Semihosting

To configure the project for semihosting, follow these steps:

1. Open **Project Properties** by right-clicking on the project in the Project Explorer.
2. Navigate to **C/C++ Build** $\rightarrow$ **Settings** $\rightarrow$ **MCU GCC Linker** $\rightarrow$ **Miscellaneous**.
3. In the **Other flags** box, add the following flags:
`-specs=rdimon.specs -lc -lrdimon`
4. In the same settings window, under **MCU GCC Linker** $\rightarrow$ **General**, ensure that the linker script is set to use `--specs=rdimon.specs`.

3 Configure Debugger for Semihosting

1. Open the **Run** $\rightarrow$ **Debug Configurations** window.
2. In the **Debugger** tab, go to **Startup** and add the following command in the **Initialization Commands** section:
`monitor arm semihosting enable`
3. Ensure that **ST-Link (OpenOCD)** is selected as the debugger.

4 Modify the syscalls.c File

If `syscalls.c` is generated in your project, it may cause conflicts. Delete the `syscalls.c` file from the `Core/Src` directory.

5 Code to Print to the Console

In your `main.c` or `main.cpp`, add the following function to support semihosting for `printf`:

```
#include <stdio.h>

#ifdef __GNUC__
#define PUTCHAR_PROTOTYPE int __io_putchar(int ch)
#else
#define PUTCHAR_PROTOTYPE int fputc(int ch, FILE *f)
#endif

PUTCHAR_PROTOTYPE
{
    ITM_SendChar(ch); // For SWV (optional, if you have ITM)
    return ch;
}

int main(void)
{
    HAL_Init();
    printf("Hello, -Semihosting!\n");

    while (1) {
        // Main loop
    }
}
```

6 Build and Debug

1. Build the project by clicking on **Project**  **Build Project**.
2. Start debugging by clicking on the **Bug icon** or selecting **Run**  **Debug**.
3. Open the **Console** view in STM32CubeIDE to see the output of the `printf` statements.

7 Final Steps

If the console doesn't show the expected output:

- Verify that semihosting is enabled in the **Debugger Startup** settings.
- Check the linker flags and make sure the correct libraries are used.
- Ensure the correct initialization of HAL and peripherals.